

Customer Lifetime Value Prediction Model

Introduction

In today's competitive retail landscape, understanding the long-term value of a customer is critical to driving profitability and strategic marketing efforts. Customer Lifetime Value (LTV) provides a quantifiable estimate of how much revenue a customer will generate over the course of their relationship with the business. In this project, we built a machine learning model to predict LTV using transaction-level data from an online retail dataset. The model helps segment customers by value and provides actionable insights for marketing and retention strategies.

Abstract

This project aims to develop a data-driven approach to estimate the Lifetime Value (LTV) of customers using historical purchase data. We used a cleaned transactional dataset containing customer IDs, invoice numbers, product details, quantity, and pricing information. After data cleaning and feature engineering, we derived key metrics such as Recency, Frequency, and Average Order Value (AOV). We then trained a regression model (XGBoost) using these features to predict monetary value as a proxy for LTV. The model performed well, with a low MAE and RMSE, and customers were segmented into Low, Medium, and High-value groups based on predicted LTV. Visualizations were used to interpret segment behavior and guide future business decisions.

Tools Used

- Python: For data analysis and modeling (Pandas, Seaborn, Scikit-learn, XGBoost)
- Google Colab: Cloud-based coding environment
- Matplotlib/Seaborn: For data visualization
- CSV: For data storage and output handling

Steps Involved in Building the Project

1. Data Cleaning

- Removed rows with missing CustomerID values
- Converted InvoiceDate to datetime format
- Created a new column $\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$

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2. Feature Engineering

- Recency: Days since last purchase
- Frequency: Number of unique purchases
- Monetary: Total spend by customer
- AOV: Average order value = Monetary / Frequency

3. Model Building

- Used XGBoost regressor to predict LTV (Monetary value)
- Split the data into training and test sets
- Evaluated model using MAE and RMSE

4. Customer Segmentation

- Applied `qcut()` on predicted LTV to classify customers into 3 segments: Low, Medium, High
- Stored results in a final CSV

5. Visualization

- Created graphs for distribution of LTV, segment-wise boxplots, scatter plots, and customer counts
- All visualizations were exported for analysis and reporting

Conclusion

This project successfully demonstrates the use of regression modeling to estimate Customer Lifetime Value from historical purchase behavior. The predictive model enables businesses to identify high-value customers, optimize retention campaigns, and focus resources on profitable segments. The LTV-based segmentation, combined with rich visual analytics, serves as a strategic tool for customer relationship management and targeted marketing. This project has enhanced my understanding of customer behavior analytics, regression modeling, and the end-to-end data science workflow.